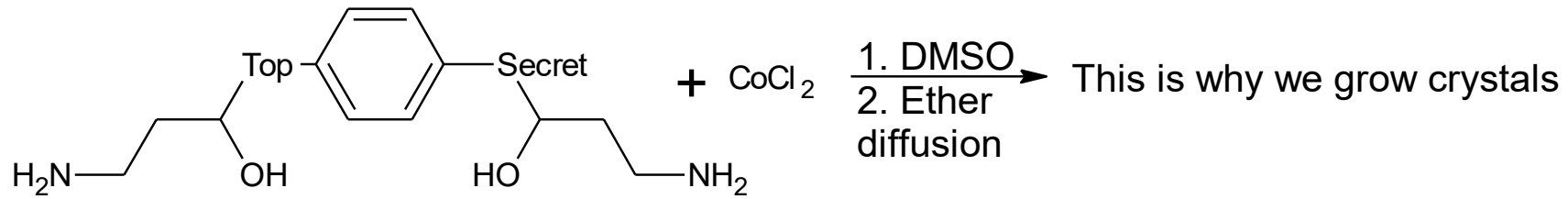
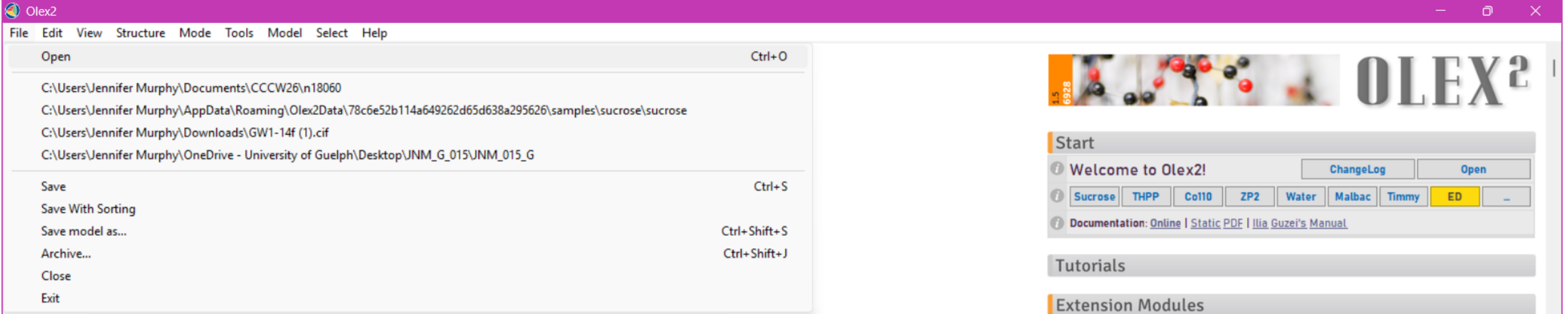


Olex2 – Getting Started

Lara Watanabe and Jen Murphy

University of Guelph



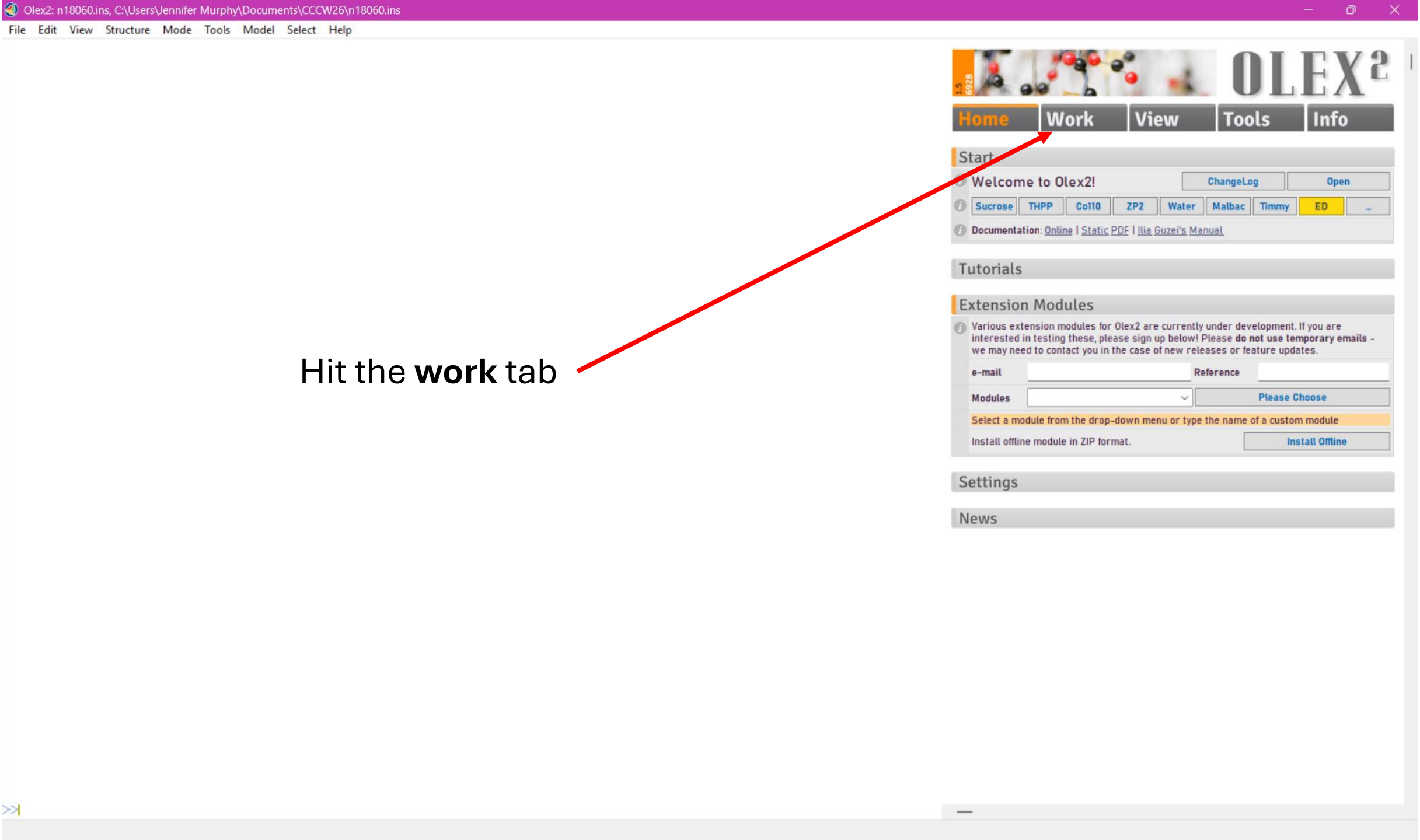


Need a file with basic information and a reflection file .hkl

Navigate to your data folder and select your .ins (sometimes .p4p)

Careful about putting your data in the subfolder of a subfolder with a long pathname

Hit the **work** tab



OLEX²

Home Work View Tools Info

Start

Welcome to Olex2! [ChangeLog](#) [Open](#)

[Sucrose](#) [THPP](#) [Co110](#) [ZP2](#) [Water](#) [Malbac](#) [Timmy](#) [ED](#) [...](#)

Documentation: [Online](#) | [Static PDF](#) | [Ilia Guzei's Manual](#)

Tutorials

Extension Modules

Various extension modules for Olex2 are currently under development. If you are interested in testing these, please sign up below! Please **do not use temporary emails** - we may need to contact you in the case of new releases or feature updates.

e-mail Reference

Modules [Please Choose](#)

Select a module from the drop-down menu or type the name of a custom module

Install offline module in ZIP format. [Install Offline](#)

Settings

News

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\i18060.ins

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

5079.0
5079
0

Solution

d min (CuKα)	0.84	I/σ(I)	32.9	Rint	7.24%	Full 132.7°	96.4
2θ=132.7°				m=9.16			
Shift	n/a	Max Peak	n/a	Min Peak	n/a	Goof	n/a

Structure n18060 loaded

Home **Work** View Tools Info

Solve  Refine  Draw  Report 

Hit this dropdown arrow, **NOT** the word “Solve”!

Toolbox Work

Labels ☐ Label H ☐ No symm

C H Cl Co O S ...  

       Z=

MAP Diff

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

Finishing

History

Select

Naming

Sorting

Select the program that you want to use to solve your structure. ShelxT has become very popular, and we will use it today, but there are other powerful ways to solve structures, especially if you are encountering problems.

Sometimes you may want to look at other possible spacegroups (you won't have Rigaku's Explain as a possibility; Bruker users could also do this in XPREP)

n18060 P1
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.ins
C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
 b = 9.4275(2) β = 81.395(2)° Z' = 1
 c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

d min (CuKα) 0.84 I/σ(I) 32.9 Rint 0.0916 7.24% Full 132.7° 96.4
 2θ=132.7°

Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Structure n18060 loaded

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

	R1	Rweak	Alpha S	sAbs Orientation	n Space	group F
ShelXT	0.103	0.016	0.051 0	00 as input	P-1	---
	0.219	0.021	0.000 0	00 as input	P1	---

Program: ShelXT (selected) Method: Intrinsic Phasing
 Reflections: Thu May 14 13:28:02 2026
 Composition: 25.0 Å³ Z = 2 Z' = 1
 Space Group: [Suggest SG](#) P-1
 Solution Settings Extra

Toolbox Work

Labels: ☐ Label H ☐ No symm Hide
 C H Cl Co O S ... Add H

Once you have selected your method for solving the structure, hit **“Solve”**

This is what you will see if you hit ‘suggest SG’

3. Cell centering test

Centering	Strong	I/Count	Count	Weak	I/Count	Count
P	-		0	-		0
A	316.75(0.30)		3977	317.70(0.29)		3982
B	315.31(0.28)		3971	319.14(0.31)		3988
C	308.11(0.29)		3989	326.38(0.30)		3970
F	305.70(0.39)		1989	321.07(0.25)		5970
I	317.66(0.30)		3986	316.79(0.29)		3973
R	321.59(0.36)		2660	315.03(0.26)		5299

Chosen lattice(s): P

Current space group: P-1 (P -1) #2

Possible space groups:

Noncentrosymmetric:

P1 (#1, Laue class -1, Point group 1)

Centrosymmetric:

P-1 (#2, Laue class -1, Point group -1)

>>|

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.ins

C₁₂H₄₂Cl₄Co₂O₆S₆

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

Solution
d min (CuKα) 0.84 I/σ(I) 32.9 R_{int} 0.0790
2θ = 132.7° m = 9.16 7.24% Full 132.7° 96.4
Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Structure n18060 loaded

Home Work View Tools Info
Solve Refine Draw Report

ShelXT	R1	Rweak	Alpha S	sAbs Orientation	n Space	group F
0.103	0.016	0.051 0	00 as input	P-1	---	
0.219	0.021	0.000 0	00 as input	P1	---	

Program ShelXT **Method** Intrinsic Phasing

Reflections n18060.hkl Thu May 14 13:28:02 2026

Composition C12 H42 Cl4 Co2 O6 S6 25.0 Å³ Z = 2 Z' = 1

Space Group **Suggest SG** P-1

Solution Settings Extra

Toolbox Work

Labels ☐ Label H ☐ No symm **Hide**

C H Cl Co O S ... **Add H**

QC QH HX QH HX Z = 1 **OK**

Fix all **AFIX** **Free XYZ** **Free ADP** **Clear HFIX** **Free H ADP**

MAP Diff **Show Map** **Map Settings**

Modelling and Disorder Tools

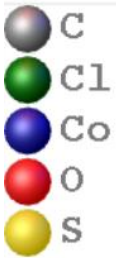
Peak & Uiso Sliders

Growing

Finishing

History

Select



If you want more or less lines type 'lines +number of lines you want' in the command line. Shown here is 'lines 20'

ShelXT Results

Structure solution: 0.286 secs

1 Centrosymmetric and 1 non-centrosymmetric space groups evaluated

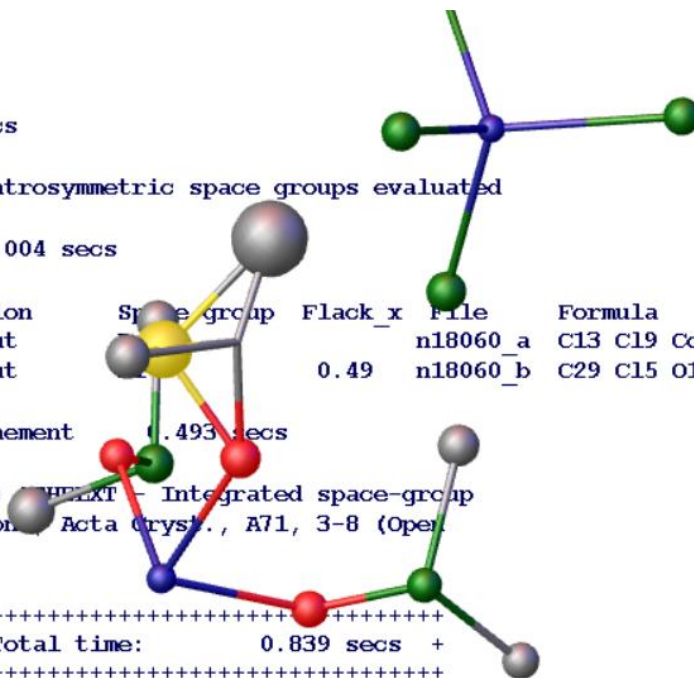
Space group determination: 0.004 secs

R1	Rweak	Alpha	SysAbs	Orientation	Space group	Flack_x	File	Formula
0.103	0.016	0.051	0.00	as input			n18060_a	C13 Cl9 Co2 O6 S
0.219	0.021	0.000	0.00	as input		0.49	n18060_b	C29 Cl5 O12 S4

Assign elements and isotropic refinement 0.493 secs

Please cite: G.M. Sheldrick (2015) SHELXT - Integrated space-group and crystal-structure determination, Acta Cryst., A71, 3-8 (Open Access) if SHELXT proves useful.

++++++
 + SHELXT finished at 14:11:59 Total time: 0.839 secs +
 ++++++



n18060 P1

C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
 b = 9.4275(2) β = 81.395(2)° Z' = 1
 c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

d min (CuKα) 2θ = 132.7° I/σ(I) 32.9 Rint 0.16 7.24% Full 132.7° 96.4
 Shift n/a Max Peak n/a Min Peak n/a GooF n/a

Running ShelXT

Home	Work	View	Tools	Info
Solve	Refine	Draw	Report	

ShelXT	R1	Rweak	Alpha S	sAbs Orientation	n Space	group F
	0.103	0.016	0.051 0	00 as input	P-1	---
	0.219	0.021	0.000 0	00 as input	P1	---

Program	ShelXT	Method	Intrinsic Phasing
Reflections	n18060.hkl		Thu May 14 14:11:59 2026
Composition	C12 H42 Cl4 Co2 O6 S6	25.0 Å³	Z = 2 Z' = 1
Space Group	Suggest SG		P-1
Solution Settings Extra			

Toolbox Work

Labels		☐ Label H	☐ No symm	Hide
	C H Cl Co O S ...	Add H		
	Q to OH H X Q H	Z= 1	OK	
	Fix all	AFIX	Free XYZ	Free ADP
	Clear HFIX	Free H ADP		
MAP	Diff	Show Map	Map Settings	
Modelling and Disorder Tools				
Peak & Uiso Sliders				
Growing				
Finishing				

History

Select

Does this solution make sense? Go back to your reaction.

A few things to try:

Fn F4

Ctrl T

Type “grow”

Type “fuse”

Go to “**Refine**” arrow
(Don’t hit “Refine” yet!)

The screenshot displays the Olex2 software interface for file n18060.res. The top section shows the chemical formula $C_{12}H_{42}Cl_4Co_2O_6S_6$ and various unit cell parameters (a, b, c, α , β , γ , Z, V). It also displays R-factors: $R_1 = 11.72\%$ and $wR_2 = 33.96\%$. A table of data follows:

d min (CuK α)	2 θ =132.7°	0.84	I/ σ (I)	32.9	Rint m=9.16	7.24%	Full 132.7°	96.4
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835	

The 'Refinement finished' message is shown. The 'Work' tab is active, and the 'Refine' button is highlighted with a red arrow. Below the main interface, the 'Toolbox Work' section is visible, containing various tools like 'Labels', 'Add H', 'Fix all', 'AFIX', 'Free XYZ', 'Free ADP', 'Clear HFIX', 'Free H ADP', 'MAP', 'Show Map', 'Map Settings', 'Modelling and Disorder Tools', 'Peak & Uiso Sliders', 'Growing', and 'Finishing'. At the bottom, there are sections for 'History', 'Select', and 'Naming'.

Disagree with the preliminary atom assignments?

Click on atoms you would like to reassign, and click on the identity that you think they should be.

(The “...” option will allow you to select atom types that were not included in your original formula.)

Do you see atoms that shouldn't be there? Click on them and hit “Delete” on your keyboard.

Note: Sometimes it can be tricky to select atoms (like in this case.) You can also select bonds and delete those (this makes selecting the atom easier.) You can also hold down Shift and your left mouse button to draw a box around the atoms you want to select.

The screenshot shows the Olex2 interface for file n18060.res. The top panel displays the chemical formula C12H42Cl4Co2O6S6 and various refinement statistics. A red arrow points from the text 'Click on atoms you would like to reassign' to the 'C' button in the 'Toolbox Work' section, which is used for selecting atom types.

a = 8.9018(2)		α = 81.945(2)°		Z = 2	42.7	R ₁	11.72 %
b = 9.4275(2)		β = 81.395(2)°		Z' = 1			
c = 18.2484(4)		γ = 89.429(1)°		V = 1499.16(6)	5079	wR ₂	33.96 %

d min (CuKα)	0.84	I/σ(I)	32.9	Rint	7.24%	Full 132.7°	96.4
2θ = 132.7°				m = 9.16			
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	Goof	2.835

Refinement finished

Home Work View Tools Info

Solve Refine Draw Report

Program: ShelXL, L.S., Client, Cycles: 10, Peaks: 59

hkl file: n18060.hkl, Thu May 14 14:43:48 2026

Weight: .100 | .112, .000 | 49.479, EXTI, SWAT, No ACTA

Use Mask: Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: [C] [H] [Cl] [Co] [O] [S] ... Add H

QC to QH, Hx, H, H, H, H, Z' = 1, OK

Fix all, AFIX, Free XYZ, Free ADP, Clear HFIX, Free H ADP

MAP: Diff, Show Map, Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

IN OLEX2 THERE ARE MANY, MANY WAYS TO ACHIEVE THE SAME GOAL! SHARE YOUR TIPS! :-D

Select your refinement program (ShelXL)

Note that it has selected the reflection file with the same file name; if you have other reflection files in this folder, you can select the one you want to use.

If you feel like you are in a good place to get started with your refinement, hit “**Refine**”

n18060
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2
b = 9.4275(2) β = 81.395(2)° Z' = 1
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

d min (CuKα) 0.84 I/σ(I) 32.9 Rint m=9.16 7.24% Full 132.7° 96.4
Shift -15.000 Max Peak 10.3 Min Peak -2.7 Goof 2.835

Refinement finished

Home Work View Tools Info

Solve Refine Draw Report

Program ShelXL LS. Client Cycles 10 Peaks 59

hkl file n18060.hkl Thu May 14 14:43:48 2026

Weight .100 | .112 .000 | 49.479 EXTI SWAT No ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels ☐ Label H ☐ No symm Hide

C H Cl Co O S ... Add H

QC to QH Hx QH H Z' = 1 OK

Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP

MAP Diff Show Map Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

How are things looking?

Ctrl T

Ctrl Q

Hover over a Q peak.

Scroll down on your mouse and the smallest peaks will start to be hidden.

If you would like to find your largest Q peak without hiding the others:

1. Intensity of the Q peak colour corresponds to size
2. Hitting this will highlight the maximum peak. What's up with that largest peak?
3. "Fn F3" will label everything. Those labels are terrible.

The screenshot displays the Olex2 software interface. The main window shows the title 'n18060' and the file path 'C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res'. The chemical formula is $C_{12}H_{42}Cl_4Co_2O_6S_6$. The unit cell parameters are listed as $a = 8.9018(2)$, $b = 9.4275(2)$, $c = 18.2484(4)$, $\alpha = 81.945(2)^\circ$, $\beta = 81.395(2)^\circ$, $\gamma = 89.429(1)^\circ$, $Z = 2$, $Z' = 1$, and $V = 1499.16(6)$. The refinement statistics are $R_1 = 11.72\%$ and $wR_2 = 33.96\%$. The data collection parameters are $d \text{ min (CuK}\alpha) = 0.84$, $2\theta = 132.7^\circ$, $I/\sigma(I) = 32.9$, $R_{int} = 7.24\%$, $R_{\sigma} = 9.16$, $\text{Full } 132.7^\circ = 96.4$, $\text{Shift} = -15.000$, $\text{Max Peak} = 10.3$, $\text{Min Peak} = -2.7$, and $\text{Goof} = 2.835$. The refinement status is 'Refinement finished'.

The Toolbox Work panel is visible, showing the 'Labels' section with a list of atoms: C, H, Cl, Co, O, S. The 'Add H' button is highlighted. The 'Z' value is set to 1. The 'Fix all' button is also visible.

When you feel confident that all non-hydrogen atoms are in your model, but before you attempt to treat any disorder, you should give your atoms sensible names.

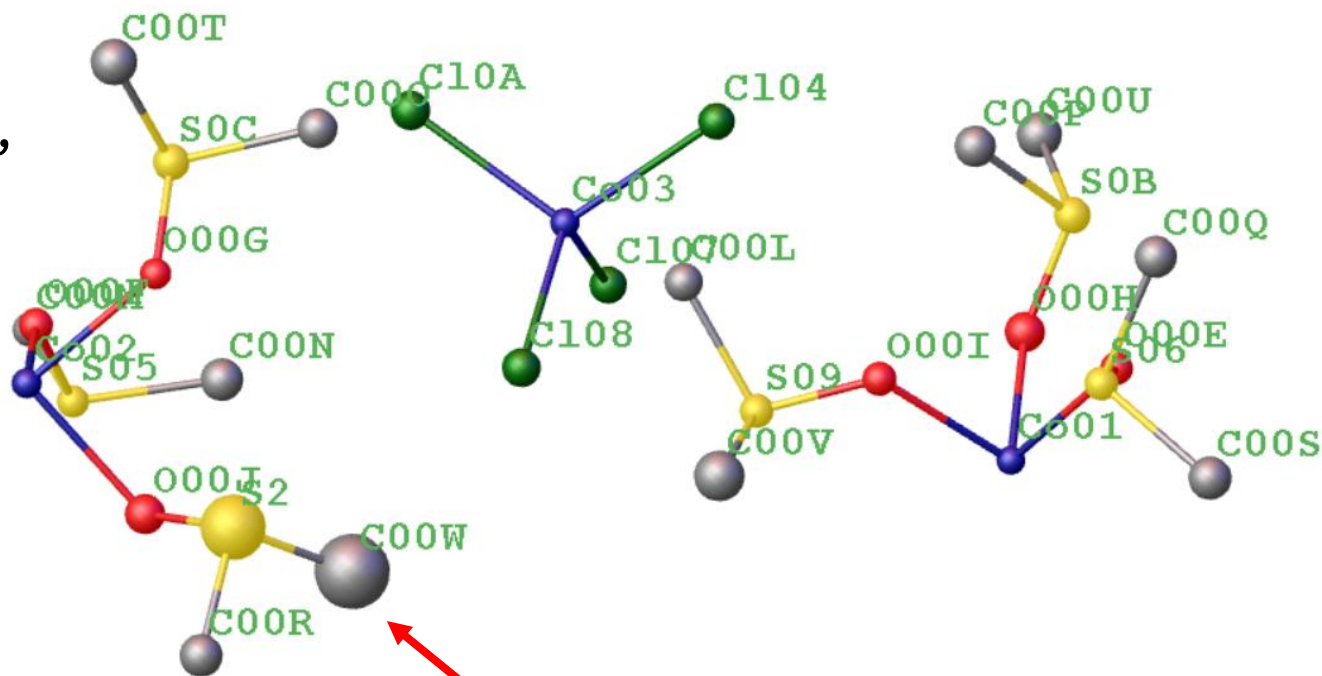
“Ctrl Q” to hide your Q peaks

Click on your cobalt atoms in the order that you would like to number them

At the prompt, type “Name 1”

Repeat for all atom types

Note: You don't have to start with 1 as your first number. For example, if you want to switch just part of a numbering sequence, select the atoms that are involved, and the number you want to start with.



If I suspect an atom, or group, to be involved in disorder, and if it makes sense to do so, I leave it for the last number.



Names okay? “**Refine**”

So far we’ve been refining everything isotropically, but we’ll want to switch to anisotropic refinement.

You can select **ACTA** to create a .cif

Then, hit the anisotropic ellipsoid to change how each atom is refined.

Everything okay so far? Hit “**Refine**”

+++++

Atom order after sorting:

```
Co1 S1 S2 S3 O1 O2 O3 C1 C2 C3 C4 C5 C6 Co2 S4 S5 S6 O4 O5 O6 C7 C8 C9 C10 C11
C12 Co3 C10A C104 C107 C108 Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15
Q16 Q17 Q18 Q19 Q20 Q21 Q22 Q23 Q24 Q25 Q26 Q27 Q28 Q29 Q30 Q31 Q32 Q33 Q34 Q35
Q36 Q37 Q38 Q39 Q40 Q41 Q42 Q43 Q44 Q45 Q46 Q47 Q48 Q49 Q50 Q51 Q52 Q53 Q54 Q55
Q56 Q57 Q58 Q59
```

The screenshot shows the Olex2 software interface. At the top, the title bar reads "n18060" and the file path is "C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res". The main window displays the chemical formula $C_{12}H_{42}Cl_4Co_2O_6S_6$ and various refinement statistics. A red box highlights the "Refine" button in the top navigation bar, and a red arrow points to the "ACTA" option in the dropdown menu. Another red arrow points to the "Anisotropic ellipsoid" icon in the "Toolbox Work" panel. The "Refinement finished" section shows the following data:

d min (CuKα)		I/σ(I)		Rint		Full 132.7°	
0.84	32.9	7.24%	96.4				
Shift	-15.000	Max Peak	10.3	Min Peak	-2.7	GooF	2.835

The "Refinement finished" section also shows the following data:

a = 8.9018(2)		α = 81.945(2)°		Z = 2	
b = 9.4275(2)	β = 81.395(2)°	Z' = 1			
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)			

The "Toolbox Work" panel shows the "Labels" section with a dropdown menu set to "No ACTA". The "Anisotropic ellipsoid" icon is highlighted. The "Refinement Settings Extra" section shows the "ACTA" option selected. The "History" section shows the "Refine" button selected.

Now how do things look?

How are your refinement values?

Where is that max peak?

Look at your Q peaks; do they make you think about hydrogen?

You can add hydrogen atoms into calculated positions by hitting the “**Add H**” button. “**Refine**”

If your data is good enough, and especially if you have hydrogen atoms involved in H-bonds, you will likely want to introduce them in their difference map positions and refine them (possibly with the use of restraints.)

The screenshot displays the Olex2 software interface for file **n18060.res**. The top panel shows the chemical formula **C₁₂Cl₄Co₂O₆S₆** and various refinement statistics. A red arrow points from the text 'How are your refinement values?' to the **R₁** value of **9.38 %**. Another red arrow points from 'Where is that max peak?' to the **Max Peak** value of **5.7**. A blue arrow points from 'Look at your Q peaks; do they make you think about hydrogen?' to the **Q to H** button in the **Toolbox Work** panel. The **Toolbox Work** panel also features an **Add H** button, which is highlighted by a red arrow from the text 'You can add hydrogen atoms into calculated positions by hitting the “Add H” button. “Refine”'. The interface includes tabs for **Home**, **Work**, **View**, **Tools**, and **Info**, and a **Solve** section with options for **Program**, **hkl file**, **Weight**, and **Refinement Settings Extra**.

a = 8.9018(2)		α = 81.945(2)°		Z = 2	
b = 9.4275(2)		β = 81.395(2)°		Z' = 1	
c = 18.2484(4)		γ = 89.429(1)°		V = 1499.16(6)	

d min (CuKα)	2θ = 132.7°	0.84	I/σ(I)	32.9	Rint	m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.364	Max Peak	5.7	Min Peak	-4.8	GooF	2.481		

Warning: 2 atoms may be split and 0 atoms NPD

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL, LS, Client, Cycles: 10, Peaks: 59

hkl file: n18060.hkl, Thu May 14 14:43:48 2026

Weight: .100 | .083, .000 | 39.551, EXTI, SWAT, ACTA

Use Mask: Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: [C] [Cl] [Co] [O] [S] ... [Add H] [OK]

Fix all, AFIX, Free XYZ, Free ADP, Clear HFIX, Free H ADP

MAP: Diff, Show Map, Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

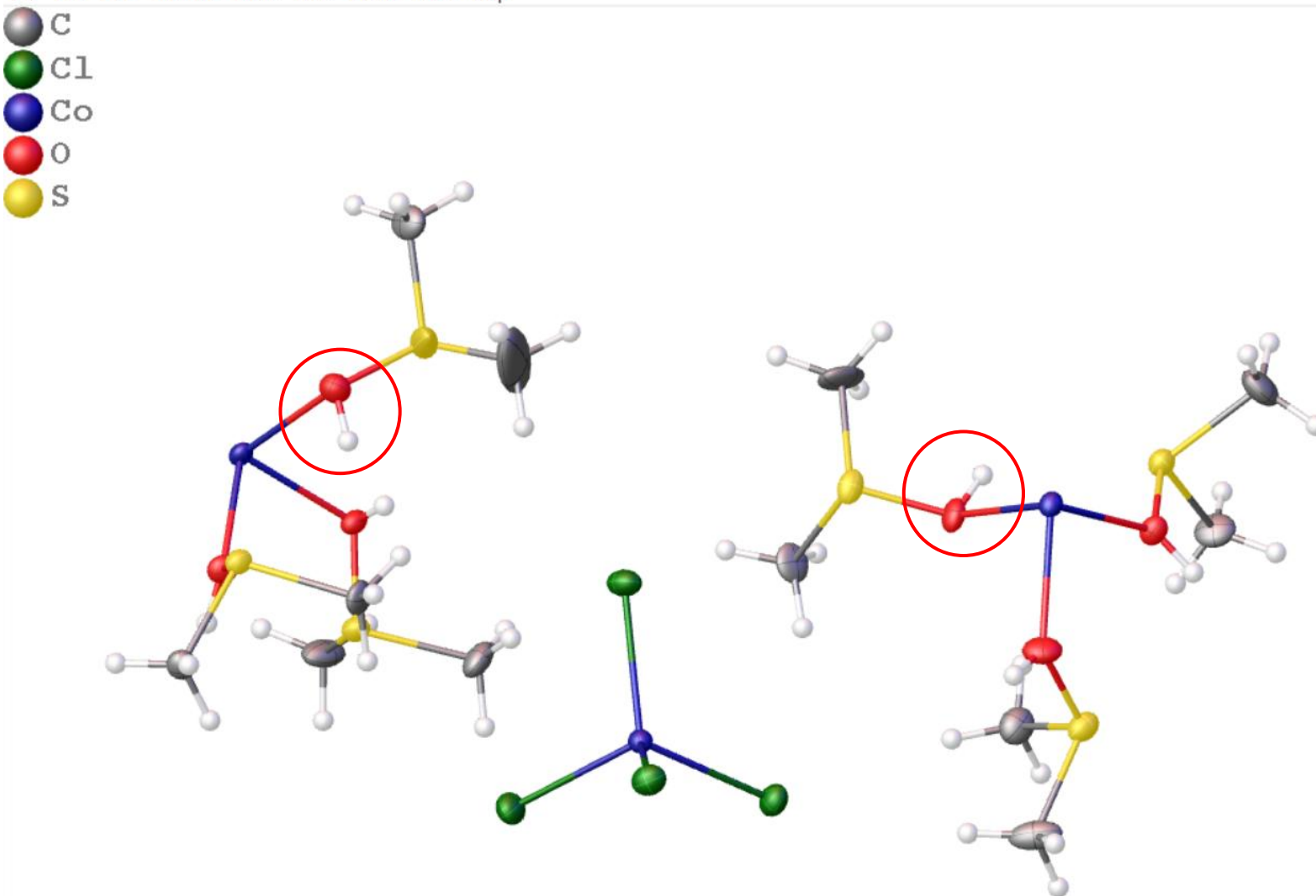
Growing

Finishing

History

Select

Naming



All the O atoms of DMSO have O atoms, that doesn't make sense.

n18060

C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C₁₂Cl₄Co₂O₆S₆

a = 8.9018(2)

b = 9.4275(2)

c = 18.2484(4)

α = 81.945(2)°

β = 81.395(2)°

γ = 89.429(1)°

Z = 2

Z' = 1

V = 1499.16(6)

18.5

5079

274

R₁

wR₂

9.38 %

29.65 %

d min (CuKα)

2θ = 132.7°

0.84

I/σ(I)

32.9

Rint

m = 9.16

7.24%

Full 132.7°

96.4

Shift

0.364

Max Peak

5.7

Min Peak

-4.8

Goof

2.481

Warning: 2 atoms may be split and 0 atoms NPD

Home

Work

View

Tools

Info

Solve

Refine

Draw

Report

Program

ShelXL

LS.

Client

Cycles

10

Peaks

59

hkl file

n18060.hkl

Thu May 14 14:43:48 2026

Weight

.100 | .083

.000 | 39.551

EXTI

SWAT

ACTA

Use Mask

Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels

C Cl Co O S ...

Add H

Q to C

Q to H

H to X

C to H

H to X

X to O

Z' = 1

OK

Fix all

AFIX

Free XYZ

Free ADP

Clear HFIX

Free H ADP

MAP

Diff

Show Map

Map Settings

Modelling and Disorder Tools

Peak & Uiso Sliders

Growing

Finishing

History

Select

Naming

1. Should all those hydrogens be there? Go back to your reaction conditions and think about charge balance.
2. Hydrogen atoms can “mask” disorder by mopping up electron density.

Go back and delete any hydrogen atoms that are relevant to these two points (click on them and hit “**Delete**” on your keyboard.)

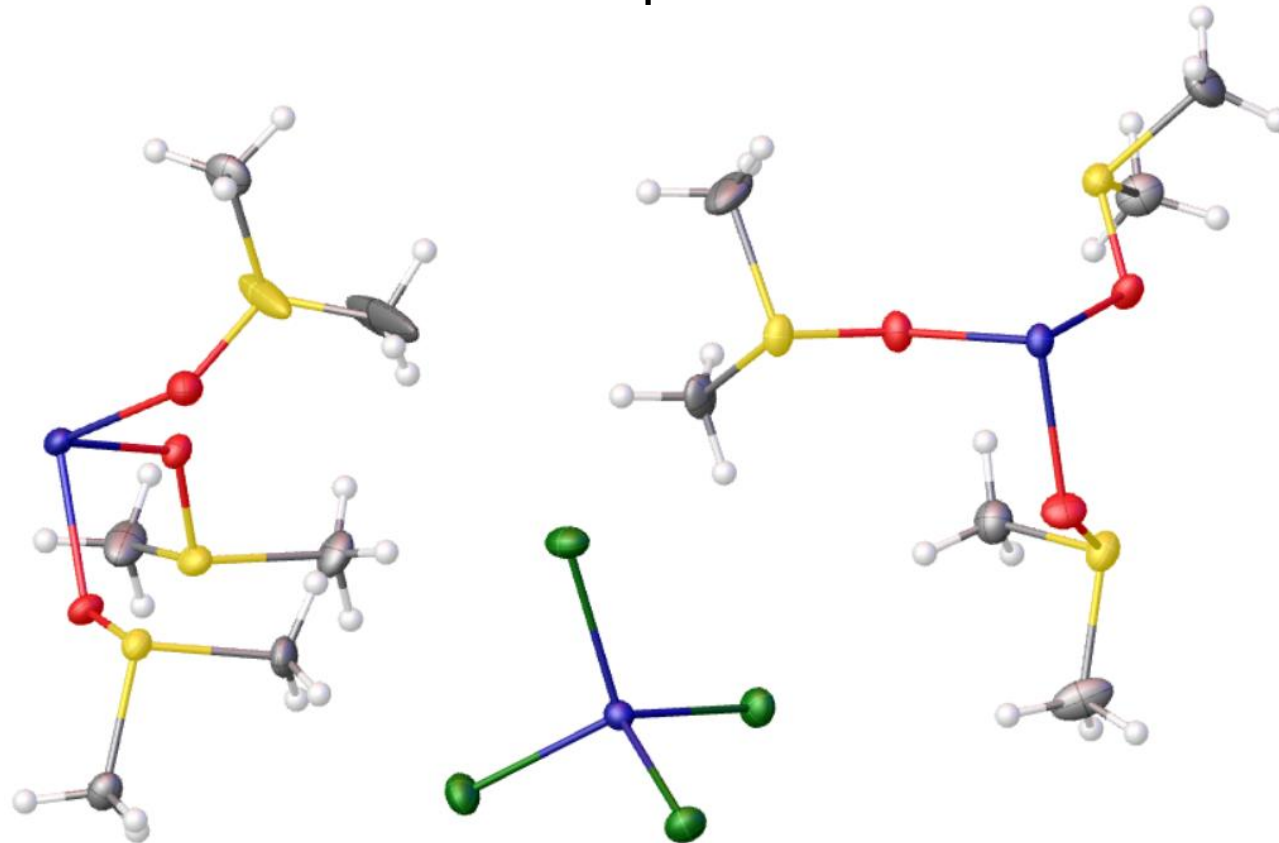
“Refine”

My refinement values improved. Did yours? This wasn’t a sure thing since points 1 and 2 were likely to off-set each other (ie. Point 1 meant that there were H-atoms introduced where none existed but point 2 meant that some H-atoms might have been used to account for electron density associated with disorder.)

Before we start dealing with disorder, let’s do a few routine “things”.



After deleting the H atoms in odd positions, the refinement statistics improved!



n18060 P1
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C₁₂Cl₄Co₂O₆S₆H_{0.5}

a = 8.9018(2) α = 81.945(2)° Z = 2
 b = 9.4275(2) β = 81.395(2)° Z' = 1
 c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6)

17.8
 5079
 286

R₁ 8.45 %
wR₂ 27.60 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	R _{int} m = 9.16	7.24%	Full 132.7°	96.4
Shift	-0.240	Max Peak	5.1	Min Peak	-5.0	GooF	2.300

Cell contents from UNIT instruction and atom list do not agree | Warning: 2 atoms may be split and 0 atoms NPD

Home **Work** **View** **Tools** **Info**
Solve **Refine** **Draw** **Report**

Program: ShelXL L.S. Client Cycles: 10 Peaks: 23
 hkl file: n18060.hkl Thu May 14 14:43:48 2026
 Weight: .100 | .046 .000 | 36.887 EXTI SWAT ACTA
 Use Mask: Use a solvent mask (smtbx.mask or SQUEEZE)
 Refinement Settings Extra

Toolbox Work

Labels: Label H No symm Hide
 C Cl Co O S H ... Add H
 Q to QH H X Q H Z' = 1 OK
 Fix all AFIX Free XYZ Free ADP Clear HFIX Free H ADP
 MAP: Diff Show Map Map Settings
 Modelling and Disorder Tools
 Peak & Uiso Sliders
 Growing
 Finishing

History

Select

Naming

At the prompt type “**edit lst**”

First, the atoms are not in a sensible order.

Close the .lst and sort your atoms

First, click on “**Sorting**” (if this menu isn’t open)

Next, decide on your sort order (I’ll just use the default) and hit “**Sort**”

“**Refine**”

You can also just type ‘**sort**’ in the command line

The screenshot shows the Olex2 software interface. At the top, the title bar reads "n18060.res, C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res". Below the title bar is a menu bar with "File", "Edit", "View", "Structure", "Mode", "Tools", "Model", "Select", and "Help". The main window displays the "n18060" project. The "Work" tab is active, showing a table of refinement statistics. Below the table, the "Refinement finished" message is displayed. The "Tools" menu is open, and the "Sorting" option is selected. The "Sorting" submenu is visible, showing the "Sort order" dropdown set to "Part", the "Label" dropdown set to "Z", and the "None" dropdown set to "None". The "Moieties" section is also visible, with a "Sort" button at the bottom right. A red arrow points from the "Sort" button in the "Sorting" submenu to the "Sort" button in the "Moieties" section.

n18060 PI

C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H42Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)° Z = 2 42.7
b = 9.4275(2) β = 81.395(2)° Z' = 1 5079 **11.72 %**
c = 18.2484(4) γ = 89.429(1)° V = 1499.16(6) 119 **33.96 %**

d min (CuKα) 0.84 I/σ(I) 32.9 Rint m=9.16 7.24% Full 132.7° 96.4
Shift -15.000 Max Peak 10.3 Min Peak -2.7 GooF 2.835

Refinement finished

Home **Work** View Tools Info

Solve **Refine** Draw Report

Program ShelXL L.S. Client Cycles 10 Peaks 59

hkl file n18060.hkl Thu May 14 14:43:48 2026

Weight ☐ .100 | .112 .000 | 49.479 EXTI ☐ SWAT ☐ **ACTA**

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History

Select

Naming

Sorting

Sort order Part Z Label None

Moieties From sort Treat H atoms independently ☐

Specific order

Atoms Reorder ☐ Move to first ☐

Moieties **Sort**

Are you up-dating your weighting? (I've been waiting until I deal with my disorder, because I know I don't yet have my best model.

What will “Info” reveal?

n18060
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)* Z = 2
b = 9.4275(2) β = 81.395(2)* Z* = 1
c = 18.2484(4) γ = 89.429(1)* V = 1499.16(6)

17.8
5079
286

R_1 8.45 %
 wR_2 27.60 %

d min (CuKα) 2θ=132.7°	0.84	I/σ(I)	32.9	Rint m=9.16	7.24%	Full 132.7°	96.4
Shift	0.000	Max Peak	5.0	Min Peak	-5.0	GooF	2.300

Warning: 2 atoms may be split and 0 atoms NPD

Home Work View Tools Info

Recent Files

Anomalous Dispersion

Difference Map Peaks

Refinement Indicators

Bad Reflections

Reflection Statistics

Reflection Statistics Summary

More info in our .lst!

n18060 P1
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2) α = 81.945(2)* Z = 2
b = 9.4275(2) β = 81.395(2)* Z* = 1
c = 18.2484(4) γ = 89.429(1)* V = 1499.16(6)

17.8
5079
286

R_1 8.45 %
 wR_2 27.60 %

d min (CuKα) 2θ=132.7°	0.84	I/σ(I)	32.9	Rint m=9.16	7.24%	Full 132.7°	96.4
Shift	0.000	Max Peak	5.0	Min Peak	-5.0	GooF	2.300

Warning: 2 atoms may be split and 0 atoms NPD

Home Work View Tools Info

Solve Refine Draw Report

Program: ShelXL L.S. Client Cycles: 10 Peaks: 23

File: n18060.hkl Thu May 14 14:43:48 2026

Weight: ☐ .100 | .046 .000 | 36.975 EXTI ☐ SWAT ☒ ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History

Select

Naming

Sorting

Sort order: Part Z Label None

Moiety: From sort Treat H atoms independently ☐

Specific order

Atoms: Reorder ☐ Move to first ☐

Moieties: Sort

At the prompt type **“edit lst”**

In the text file, search for

“disagreeable”

“split”

“Highest peak”

There is so much great information in this file!

Okay, at this point, we can either model this disorder, or leave that to someone else...

What time is it?

In your main screen “Ctrl T” until you have a blank background and “Ctrl Q” to show your Q-peaks, but scroll down on your mouse so that only the two highest ones are displayed.

Open Toolbox Work by clicking on it

There are multiple ways to treat disorder in OLEX2; you can click on the “i” to get info about the various settings that you can apply.

```
+++++
+ n18060      finished at 09:11:19   Total elapsed time:    0.25 secs +
+++++
```

Note: Q-peaks are invisible
The following atom(s) may be split:
S6 C12

>>|

The screenshot displays the Olex2 software interface. The top section shows the file path and the chemical formula $C_{12}H_{36}Cl_4Co_2O_6S_6$. Below this, a table of unit cell parameters is shown:

a = 8.9018(2)	α = 81.945(2)°	Z = 2	17.8	R_1	8.50 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	wR_2	27.69 %
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	285		

Below the unit cell parameters, a table of refinement statistics is shown:

d min (CuKα)	0.84	I/σ(I)	32.9	Rint	7.24 %	Full 132.7°	96.4
2θ = 132.7°				m = 9.16			
Shift	0.010	Max Peak	5.1	Min Peak	-5.0	Goof	2.307

A red box highlights a warning message: "Cell contents from UNIT instruction and atom list do not agree | Warning: 2 atoms may be split and 0 atoms NPD".

The interface includes a menu bar (File, Edit, View, Structure, Mode, Tools, Model, Select, Help) and a toolbar with icons for Solve, Refine, Draw, and Report. The "Toolbox Work" panel is open, showing various settings for labels, refinement, and disorder treatment.

The "Toolbox Work" panel includes sections for:

- Labels: A dropdown menu and checkboxes for "Label H" and "No symm".
- Refinement: Buttons for "Fix all", "AFIX", "Free XYZ", "Free ADP", "Clear HFIX", and "Free H ADP".
- MAP: A dropdown menu for "Diff" and buttons for "Show Map" and "Map Settings".
- Modelling and Disorder Tools: Buttons for "Fit", "Split", and "PART", along with dropdown menus for "auto" and "FVAR".
- FVAR: A dropdown menu for "2.0".
- Peak & Uiso Sliders: A section for adjusting peak and Uiso values.
- Growing: A section for growing the model.
- Finishing: A section for finishing the model.

The "History" panel is visible at the bottom of the interface.

I recommend deleting the H-atoms on the C that is involved in the disorder.

I also like to make these atoms isotropic first.

Select the C and S that are involved in disorder, and hit “**Split**”, leaving the other settings as shown.

Hover your mouse over the group you want to split, and hold “**Ctrl**” and your left mouse button while moving your mouse around until the second component fits the two highest Q peaks

“**Refine**”

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2	17.8	R_1	8.50 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	wR_2	27.69 %
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	285		

d min (CuK α) 2 θ = 132.7°	0.84	I/ σ (I)	32.9	Rint m = 9.16	7.24 %	Full 132.7°	96.4
Shift	0.010	Max Peak	5.1	Min Peak	-5.0	Goof	2.307

Cell contents from UNIT instruction and atom list do not agree | Warning: 2 atoms may be split and 0 atoms NPD

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL | L.S. | Client | Cycles: 10 | Peaks: 26

hkl file: n18060.hkl | Thu May 14 14:43:48 2026

Weight: ☐ .100 | .049 | .000 | 37.389 | EXTI ☐ SWAT ☐ ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: | ☐ Label H ☐ No symm | Hide

C H Cl Co O S ... | Add H

QC QH H H H H H | Z' = 1 | OK

Fix all | AFIX | Free XYZ | Free ADP | Clear HFIX | Free H ADP

MAP: Diff | Show Map | Map Settings

Modelling and Disorder Tools

Fit | Split | with SAME | PART auto | FVAR auto

FVAR: |

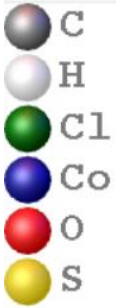
☐ Show/Hide the list of restraints No Restraints are shown 2.0

Peak & Uiso Sliders

Growing

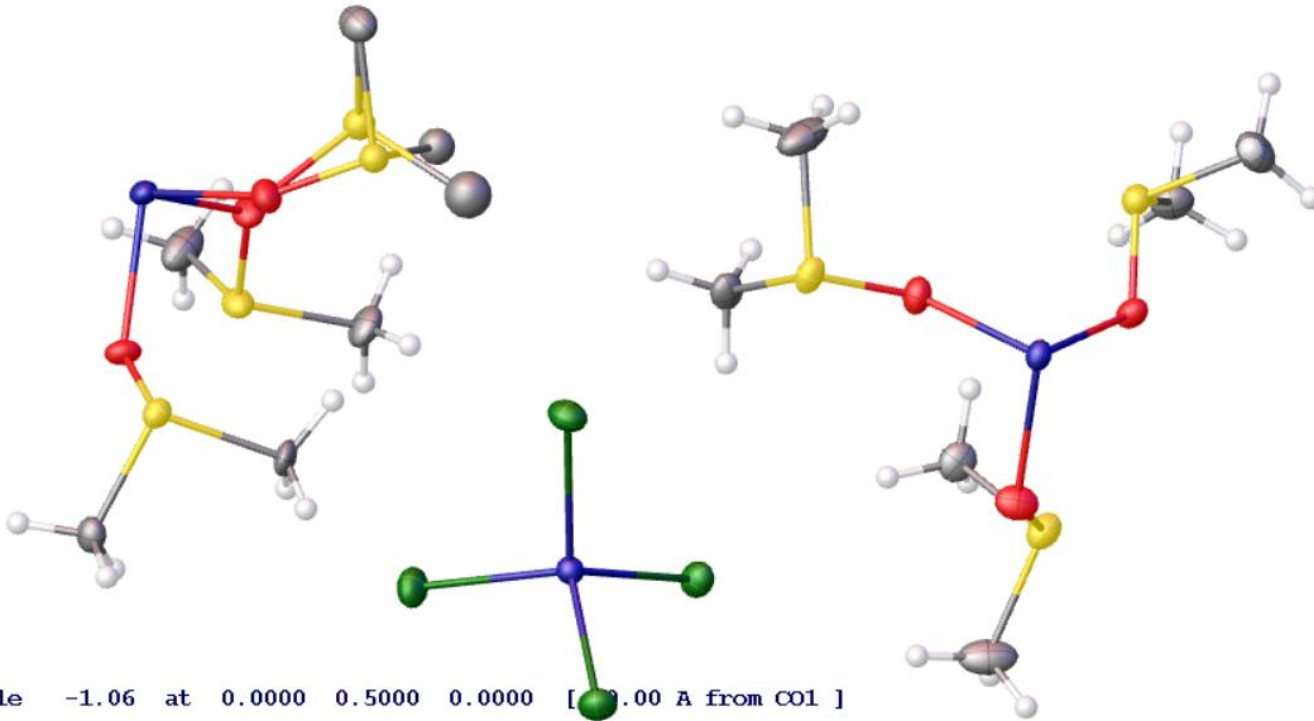
Finishing

History



My structure after ESC from the “split” setting

Hover over the split atoms



Please cite: G.M. Sheldrick (2015) "Crystal structure refinement with SHELXL", Acta Cryst., C71, 3–8 (Open Access) if SHELXL proves useful.

```

+++++
+ n18060      finished at 09:14:15  Total elapsed time:    0.25 secs +
+++++

```

Note: Q-peaks are invisible

>>|

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2	17.9	R_1	4.89 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079		
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	283	wR_2	15.34 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	-0.001	Max Peak	0.9	Min Peak	-1.1	Goof	1.250

Cell contents from UNIT instruction and atom list do not agree

Home **Work** View Tools Info

Solve **Refine** Draw Report

Program: ShelXL LS. Client Cycles: 10 Peaks: 36

hkl file: n18060.hkl Thu May 14 14:43:48 2026

Weight: ☐ .100 | .049 .000 | 5.029 EXTI ☐ SWAT ☐ ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: ☐ Label H ☐ No symm

...

Z' = 1

MAP: Diff

Modelling and Disorder Tools

Set ☒ Unique ☒ Labels

| with SAME PART auto FVAR auto

FVAR: | 21 0.654 |

☐ Show/Hide the list of restraints No Restraints are shown 2.0

Peak & Uiso Sliders

Growing

Finishing

If you aren't happy about how these two atoms were split, you can always go back in history to an earlier model.

Just click on any of the Histogram bars, and the model that corresponded to that round of least squares will pop back up on your screen!

If you just don't like how you split your atoms, and want a second try at it, hit the last bar.

n18060 PI
C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H30Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2		17.9	R_1	4.89 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1		5079		
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)		283		

wR_2 **15.34 %**

d min (CuK α) 2 θ = 132.7°	0.84	I/ σ (I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	-0.001	Max Peak	0.9	Min Peak	-1.1	Goof	1.250

Cell contents from UNIT instruction and atom list do not agree

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program ShelXL LS. Client Cycles 10 Peaks 36

hkl file n18060.hkl Thu May 14 14:43:48 2026

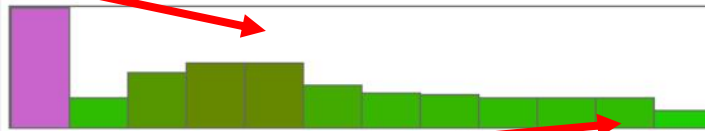
Weight ☐ .100 | .049 ☐ .000 | 5.029 EXTI ☐ SWAT ☒ ACTA

☐ **Use Mask** Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

History



ShelXL (Least Squares) - 4.89%

Scale 5

[Reload ins](#) [Create Branch](#) [Create Snapshot](#)

History Tree

Select

Naming

Sorting

You should notice that after you split your atoms they now look like they are being refined isotropically (because they are; anisotropic refinement can also “mop up” electron density, so we refine disorder isotropically first.) Notice the nice work OLEX2 did naming the second disorder component?

You should also notice that the H-atoms on the “not disordered” carbon disappeared. Why do you think that is?

Peter Mueller would tell you that you will never refine disorder without restraints. Let’s open our .ins to see what was added.

“edit .ins”

```
TITL n18060_a.res in P-1
REM Old TITL n18060_a.res in P-1
REM SHELXT solution in P-1: R1 0.103, Rweak 0.016, Alpha 0.051
REM <I/s> 0.000 for 0 systematic absences, Orientation as input
REM Formula found by SHELXT: C13 C19 Co2 O6 S
CELL 1.54184 8.9018 9.4275 18.2484 81.945 81.395 89.429
ZERR 2 0.0002 0.0002 0.0004 0.002 0.002 0.001
LATT 1
SFAC C H Cl Co O S
UNIT 24 72 8 4 12 12
SADI S6 C11 S6A C11
SADI S6 C12 S6A C12A
SADI 0.04 06 C12 06 C12A
SADI 0.04 C11 C12 C11 C12A

L.S. 10
PLAN 24
CONF
LIST 6
fmap 2
MORE -1
ACTA
BOND $H
WGHT 0.1
FVAR 0.46967 0.6548
REM <olex2.extras>
REM <HklSrc "%.\\n18060.hkl">
REM </olex2.extras>
```

Peter says it here:

<https://www.tandfonline.com/doi/pdf/10.1080/08893110802547240?needAccess=true>

We can add additional restraints and constraints.

First, open up the “**Tools**” menu

Then, open up “**Shelx Compatible Restraints**” or “**Constraints**”

You should not simultaneously refine occupancy and displacements (at least on your first round of least squares).

We can apply an **EADP** constraint to force selected displacements to be identical.

Later, we can edit these constraints to replace with **RIGU** restraints.

Shelx instructions!

http://shelx.uni-goettingen.de/shelxl_html.php

c = 18.2484(4)		γ = 89.429(1)°		V = 1499.16(6)		306	<i>wR</i> ₂	14.14 %
d min (CuKα)	0.84	I/σ(I)	32.9	R _{int} m=9.16	7.24%	Full 132.7°	96.4	
Shift	0.001	Max Peak	0.5	Min Peak	-1.1	Goof	1.154	

Refinement finished

Home Work View **Tools** Info

FragmentDB

AC7

Twinning

Images

Maps

Chemical Tools

Olex2 Constraints Restraints

Shelx Compatible Constraints

Shelx Compatible Restraints

Hydrogen Atoms

Disorder

RIGU!

<http://journals.iucr.org/a/issues/2012/04/00/pc5011/pc5011.pdf>

If everything looked stable, hit your **anisotropic button** again, and “**Refine**”. If things still look good, you will want to add H-atoms.

This time, just select the carbon atoms in the disordered group (if you hit “**Add H**” before doing this, it will add back in the O-H hydrogen atoms that you already deleted.)

Notice how the carbon atom that was not split now has six hydrogens on it? Let’s see how this makes sense by selecting to view only one “**part**” at a time.

You can also select to update your weights now.

n18060 PI
 C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)	α = 81.945(2)°	Z = 2	16.6	R_1 4.54 %
b = 9.4275(2)	β = 81.395(2)°	Z' = 1	5079	
c = 18.2484(4)	γ = 89.429(1)°	V = 1499.16(6)	306	wR_2 14.14 %

d min (CuKα) 2θ = 132.7°	0.84	I/σ(I)	32.9	Rint m = 9.16	7.24%	Full 132.7°	96.4
Shift	0.001	Max Peak	0.5	Min Peak	-1.1	Goof	1.154

Refinement finished

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

Program: ShelXL | LS. | Client | Cycles: 10 | Peaks: 24

hkl file: n18060.hkl | Thu May 14 14:43:48 2026

Weight: ☒ | .100 | .046 | .000 | 3.466 | EXTI ☐ SWAT ☐ ACTA

☐ Use Mask | Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: | ☐ Label H ☐ No symm | Hide

C H Cl Co O S ... | Add H

QC to H | H | H | H | H | H | Z = 1 | OK

Fix all | AFIX | Free XYZ | Free ADP | Clear HFIX | Free H ADP

MAP: Diff | Show Map | Map Settings

Modelling and Disorder Tools

0 | 1 | 0 | 2 | All | Set ☒ Unique ☒ Labels

Fit | Split with SAME | PART auto | FVAR auto

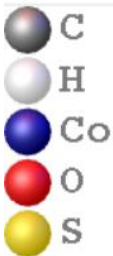
FVAR: | 21 0.655 |

☐ Show/Hide the list of restraints | No Restraints are shown | 2.0

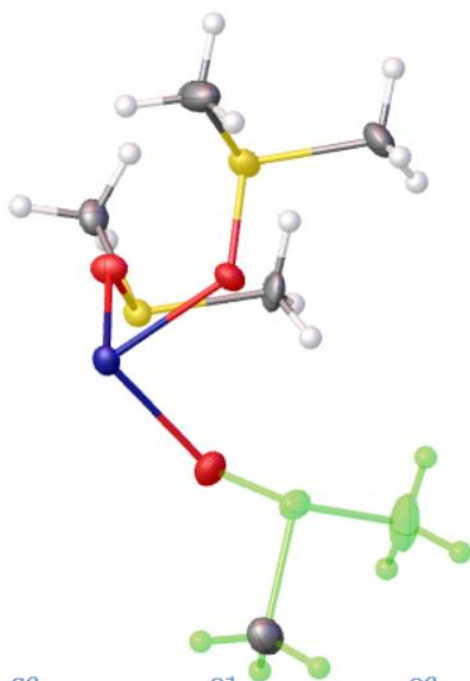
Peak & Uiso Sliders

Growing

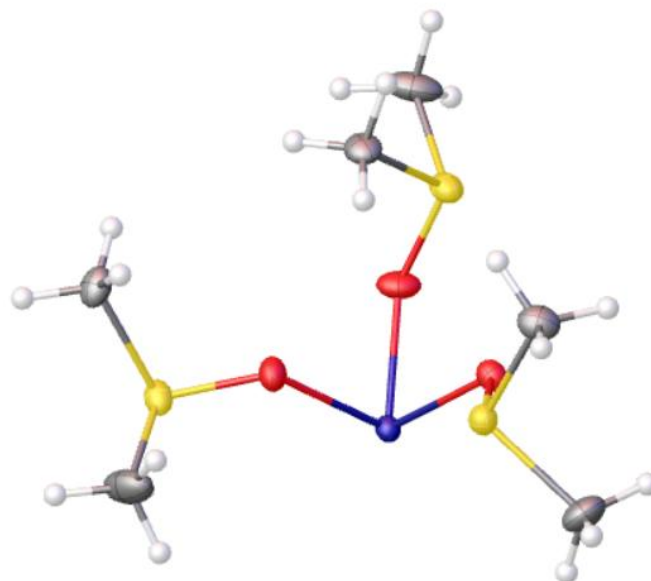
Finishing



With one part shown



C3	S2	O1	O2	H1B
H2A	H1A	O3	H2B	H3A
S1	S3	H3B	H1C	C8
O5	H8A	H8C	O6	C6
H6A	S5	H8B	C4	H4A
S4	H3C	Co2	C5	C7
H4B	H5C	H7B	C9	H6C
S6	H9A	H6B	H9B	H7A
O4	H4C	H5A	H5B	H7C



n18060

PI

C:\Users\Jennifer Murphy\Documents\CCCW26\118060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)

 α = 81.945(2) $^\circ$

Z = 2

b = 9.4275(2)

 β = 81.395(2) $^\circ$

Z' = 1

c = 18.2484(4)

 γ = 89.429(1) $^\circ$

V = 1499.16(6)

 16.6
5079
306
 R_1

4.54 %

 wR_2

14.14 %

d min (CuK α)

0.84

I/ σ (I)

32.9

Rint

m=9.16

7.24%

Full 132.7 $^\circ$

96.4

Shift

0.001

Max Peak

0.5

Min Peak

-1.1

Goof

1.154

Refinement finished

Home

Work

View

Tools

Info

Solve

Refine

Draw

Report

i

Program

ShelXL

LS.

☐ Client

Cycles

10

Peaks

24

i

hkl file

n18060.hkl

Thu May 14 14:43:48 2026

i

Weight

☒

.100 | .046

.000 | 3.466

EXTI

☐ SWAT☐ ACTA

i

☐ Use Mask

Use a solvent mask (smtbx.mask or SQUEEZE)

i

Refinement Settings Extra

Toolbox Work

i

Labels

☐ Label H☐ No symm

Hide

i

C

H

Cl

Co

O

S

...

Add H

i

Q to C

Q to H

H to X

Q to H

H to X

H to X

H to X

H to X

H to X

H to X

i

Fix all

AFIX

Free XYZ

Free ADP

Clear HFIX

Free H ADP

i

MAP

Diff

Show Map

Map Settings

i

Modelling and Disorder Tools

i

0 | 1

0 | 2

All

Set

☒ Unique☒ Labels

i

Fit

Split

with

SAME

PART

auto

FVAR

auto

i

FVAR:

| 21 0.655 |

i

☐ Show/Hide the list of restraints

No Restraints are shown

i

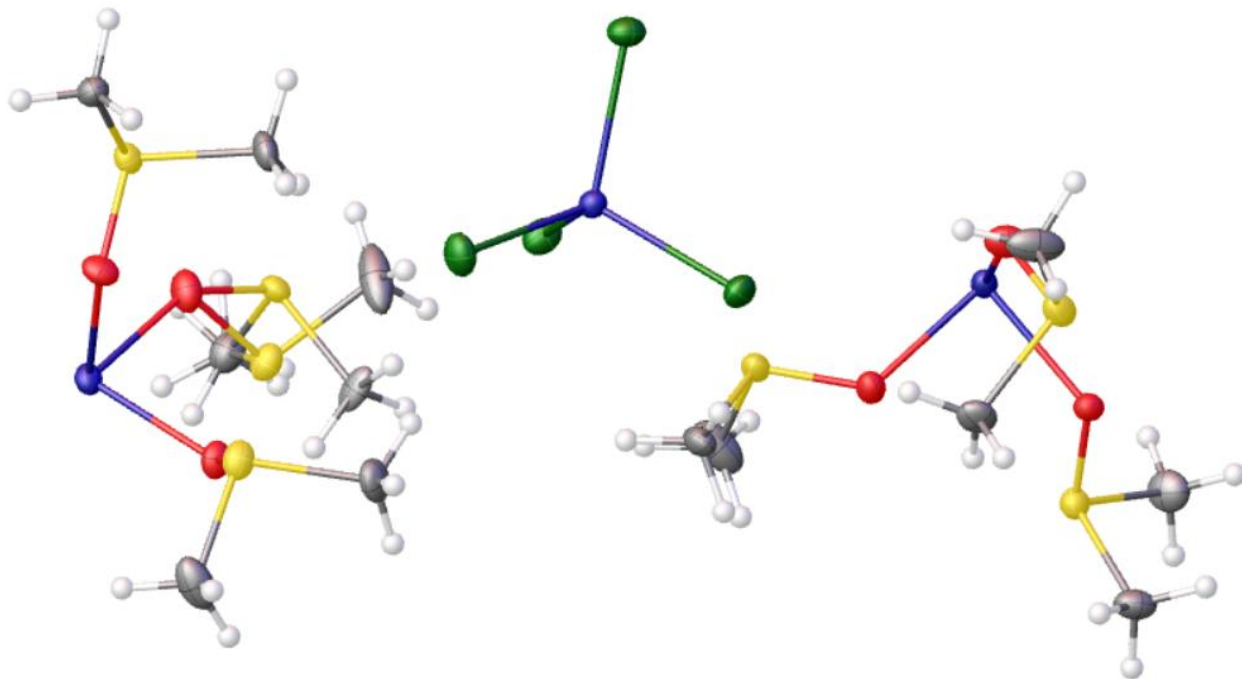
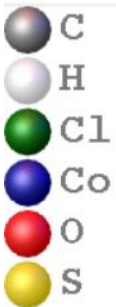
Peak & Uiso Sliders

i

Growing

i

Finishing



Deepest hole -0.39 at 0.9042 0.5221 0.4507 [1.32 Å from CO2]

Please cite: G.M. Sheldrick (2015) "Crystal structure refinement with SHELXL", Acta Cryst., C71, 3–8 (Open Access) if SHELXL proves useful.

```

+++++
+ n18060      finished at 09:21:11  Total elapsed time:    0.25 secs +
+++++

```

Note: Q-peaks are invisible

>>|

n18060

P1

C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C₁₂H₃₆Cl₄Co₂O₆S₆

a = 8.9018(2)

α = 81.945(2)°

Z = 2

b = 9.4275(2)

β = 81.395(2)°

Z' = 1

c = 18.2484(4)

γ = 89.429(1)°

V = 1499.16(6)

16.6
5078
306

R₁

4.34 %

wR₂

8.85 %

d min (CuKα)

2θ = 132.7°

Shift

-0.001

I/σ(I)

33.0

Rint

m = 9.16

7.24 %

Full 132.7°

96.3

Min Peak

-0.4

GooF

1.145

Refinement finished

Home

Work

View

Tools

Info

Solve

Refine

Draw

Report

Program

ShelXL

LS

Client

Cycles

10

hkl file

n18060.hkl

Peaks

24

Weight

.027 | .027

3.96 | 3.97

EXTI

SWAT

ACTA

Use Mask

Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels

Label H

No symm

Hide

C

H

Cl

Co

O

S

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

H to H

H to H

Q to Q

H to H

H to H

H to H

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H to H

Q to Q

H to H

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Q to Q

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Q to Q

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H to H

H to H

H to H

H to H

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Q to Q

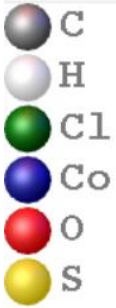
H to H

H to H

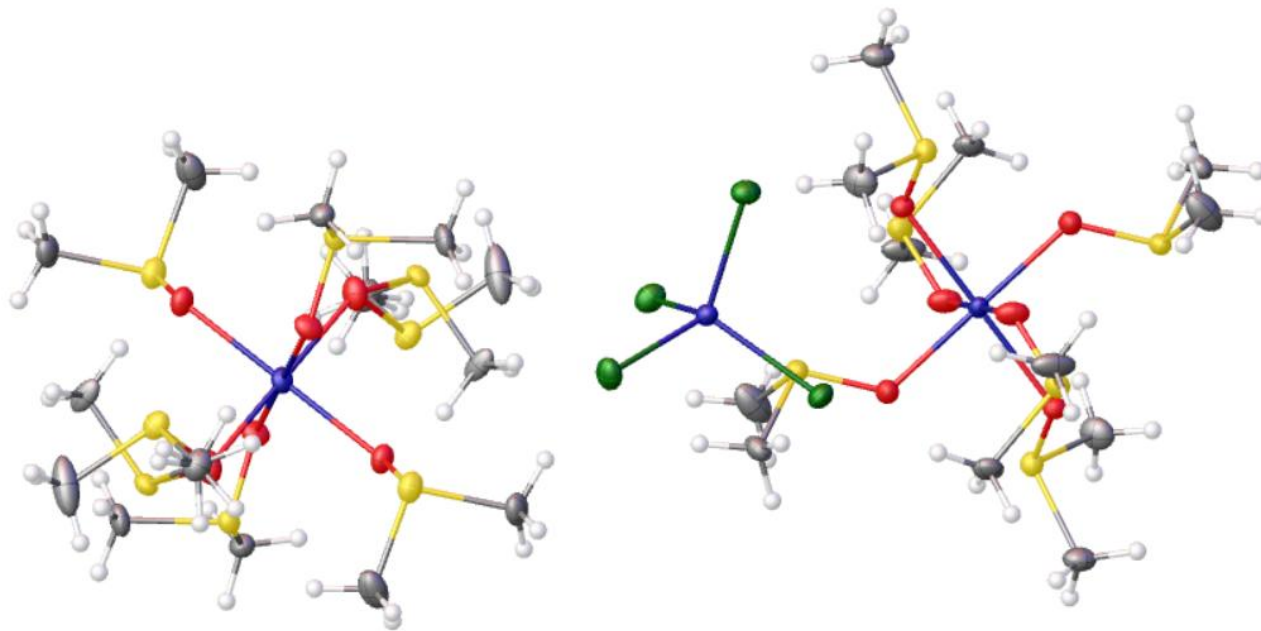
H to H

H to H

H to H



When “grown”



Deepest hole -0.39 at 0.9042 0.5221 0.4507 [1.32 Å from CO₂]

Please cite: G.M. Sheldrick (2015) "Crystal structure refinement with SHELXL", *Acta Cryst.*, C71, 3–8 (Open Access) if SHELXL proves useful.

```

+++++
+ n18060      finished at 09:21:11  Total elapsed time:    0.25 secs +
+++++

```

Note: Q-peaks are invisible

>>|

n18060

C:\Users\Jennifer Murphy\Documents\CCCW26\n18060.res

C12H36Cl4Co2O6S6

a = 8.9018(2)

b = 9.4275(2)

c = 18.2484(4)

α = 81.945(2)°

β = 81.395(2)°

γ = 89.429(1)°

Z = 2

Z' = 1

V = 1499.16(6)

16.6

5078

306

R₁

wR₂

Full 132.7°

Goof

1.145

4.34 %

8.85 %

d min (CuKα)

2θ = 132.7°

Shift

-0.001

Max Peak

0.4

Min Peak

-0.4

Refinement finished

Home

Work

View

Tools

Info

Solve

Refine

Draw

Report

Program

ShelXL

LS.

Client

Cycles

10

Peaks

24

hkl file

n18060.hkl

Thu May 14 14:43:48 2026

Weight

.027 | .027

3.96 | 3.97

EXTI

SWAT

ACTA

Use Mask

Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels

Label H

No symm

Hide

C H Cl Co O S ...

Add H

Q to Q

H to H

Q to H

H to Q

Q to Q

H to H

Q to H

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In this tutorial I have deliberately not included screenshots of my model because I didn't want to give away the answer!

I've posted this as a file that you can edit. Why don't you go back, work through it again, and add your own notes and screenshots? You can start from scratch by going to Model → Reset

The screenshot displays the Olex2 software interface for file `n18060.res`. On the left, a legend identifies atoms: C (grey), H (white), Cl (green), Co (blue), O (red), and S (yellow). A context menu is open over the model, showing options like Refine, Move, Q-->C, Tidy, ANIS All, ISOT All, ANIS Selected, ISOT Selected, Reset, and Solve. The central area shows a 3D ball-and-stick model of a complex molecule. On the right, a summary panel for **n18060** provides the chemical formula C12H36Cl4Co2O6S6 and various refinement metrics.

Unit Cell Data		Angles		Z		R-factors		Goodness	
a = 8.9018(2)	α = 81.945(2)°	b = 9.4275(2)	β = 81.395(2)°	c = 18.2484(4)	γ = 89.429(1)°	Z = 2	R_1 = 4.34 %	Full 132.7° = 96.3 %	Goof = 1.145
d min (CuK α) = 0.84	I/ σ (I) = 33.0	Rint = 7.24 %	Max Peak = 0.4	Min Peak = -0.4	Shift = -0.001	16.6 / 5078 / 306	Full 132.7° = 96.3 %	Goof = 1.145	

Refinement finished

Home | Work | View | Tools | Info

Solve | Refine | Draw | Report

Program: ShelXL | LS. | Client | Cycles: 10 | Peaks: 24

hkl file: n18060.hkl | Thu May 14 14:43:48 2026

Weight: ☒ .027 | .027 | 3.96 | 3.97 | EXTI ☐ SWAT ☐ ACTA

☐ Use Mask Use a solvent mask (smtbx.mask or SQUEEZE)

Refinement Settings Extra

Toolbox Work

Labels: | Label H ☐ No symm ☐ Hide

C H Cl Co O S ... | Add H

QC to QH | Hx | Q | H | Z = 1 | OK

Fix all | AFIX | Free XYZ | Free ADP | Clear HFIX | Free H ADP

MAP: Diff | Show Map | Map Settings

Modelling and Disorder Tools